

Fei Wu (吴飞)

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EDUCATION

- Ph.D. in Computer Science (Fully Funded), University of Exeter (Exeter, UK)** 10/2023-07/2027
- **Research Interests:** parameter-efficient fine-tuning (PEFT) of LLMs, federated foundation model fine-tuning, efficient agent memory system, algorithm-hardware co-design, and efficient AI systems
 - ★ **Scholarships:** China Scholarship Council (CSC) Scholarship, The Limetree Capital PhD Scholarship (£5,000, 2023-2027)
- M.Eng. in Electronics and Communication Engineering (Graduate recommendation & exam exemption), University of Electronic Science and Technology of China (Chengdu, China)** 09/2019-07/2022
- **Research Interests:** AI hardware accelerators - Intelligent Chips and Systems for the Internet of Things Research Group
 - **Overall Ranking:** 23/332 (Top 7%)
 - ★ **Honors:** Outstanding Graduate of Sichuan Province & Outstanding Graduate of the University
 - ★ **Scholarship:** First-Class Graduate Academic Scholarship, UESTC (¥10000, 2021)
- B.Eng. in Electronic Information Engineering, Chengdu University of Technology (Chengdu, China)** 09/2015-07/2019
- **Overall Ranking:** 1/135 (Top 1%)
 - ★ **Honors:** Outstanding Student, Chengdu University of Technology (2016-2018)
 - ★ **Scholarship:** Academic Scholarship, Chengdu University of Technology (¥4,500, 2016-2018)

WORK EXPERIENCE

- Research Assistant, University of Exeter & Institute of Applied AI and Robotics (IAAIR), USA** 07/2026-Present
- Conduct research on applying parameter-efficient fine-tuning of LLMs to smart agriculture applications
- Research Assistant, University of Exeter** 01/2025-07/2025
- Conduct research on parameter-efficient fine-tuning of LLMs and federated foundation model fine-tuning systems
- FPGA Engineer, Chengdu Ruixinxing Technology Co., Ltd.** 10/2022-07/2023
- Developed efficient algorithm-hardware co-design SLAM solutions for autonomous mobile robots

PUBLICATIONS

- [1] F. Wu, J. Hu, G. Min, S. Wang, “[Preventing Rank Collapse in Federated Low-Rank Adaptation with Client Heterogeneity](#)”, *ArXiv*, Feb 2026. (Submitted to *NeurIPS 2026*)
- [2] F. Wu, J. Hu, G. Min, S. Wang, “[PSOFT: Efficient Orthogonal Fine-Tuning with Principal Subspace Adaptation](#)”, *The Fourteenth International Conference on Learning Representations (ICLR 2026)*, Apr 2026. (CCF A)
- [3] F. Wu, J. Hu, G. Min, S. Wang, “[Adaptive Rank Allocation for Federated Parameter-Efficient Fine-Tuning of Language Models](#)”, *IEEE Transactions on Computers (IEEE TC)*, vol. 75, no. 4, pp. 1650-1663, 2026. (CCF A)
- [4] Y. Liu, F. Wu, N. Zhao et al., “[NVP: A Flexible and Efficient Processor Architecture for Accelerating Diverse Computer Vision Tasks including DNN](#)”, *IEEE Transactions on Circuits and Systems II: Express Briefs (IEEE TCAS-II)*, vol. 70, no. 1, pp. 271-275, 2023. (JCF Q1)
- [5] Z. Wang, F. Wu, F. Yu, Yurui Zhou, Jia Hu, Geyong Min, “[Federated Continual Learning for Edge-AI: A Comprehensive Survey](#)”, *ArXiv*, Nov 2024.
- [6] Y. Liu, J. Xiao, F. Wu et al., “[A Fast and Efficient FPGA-based Level Set Hardware Accelerator for Image Segmentation](#)”, *Journal of Electronics and Information Technology (JEIT)*, vol. 32, no. 6, pp. 1525-1532, 2021.
- [7] F. Wu, N. Zhao, Y. Liu et al., “[A Review of Convolutional Neural Networks Hardware Accelerators for AIoT Edge Computing](#)”, *The International Conference on UK-China Emerging Technologies (UCET)*, 2021, pp. 71-76.

OPEN-SOURCE CONTRIBUTIONS

- PSOFT Tuner Pull Request - Integrated into 🤗 Hugging Face PEFT v0.19.0 (★21.6k Stars)** 02/2026-03/2026
- Proposed principal-subspace orthogonal fine-tuning (PSOFT), achieving 18× higher parameter efficiency while avoiding out-of-memory (OOM) issues in SOTA orthogonal fine-tuning
 - Validated the effectiveness of PSOFT across GLUE, VTAB-1K, GSM8K, and Commonsense Reasoning tasks
 - Developed ~1,500 lines of production-grade code, passed the official code review, and merged the implementation into the Hugging Face PEFT main branch v0.19.0

RESEARCH & ENGINEERING PROJECTS

- SALM: Small Agriculture Language Model for Rice Pathogen Diagnosis, IAAIR** 2026-Present
- Designed a parameter-efficient fine-tuning pipeline for agricultural data and evaluated domain accuracy and reliability
 - Integrated a knowledge graph with the fine-tuned model, implemented model merging and low-bit GGUF quantization, and deployed the system on mobile devices using llama.cpp
 - ★ Expected Outcome: A mobile AI assistant prototype for rice pathogen diagnosis & several research publications

- WAMS: Workload-aware Efficient Agent Memory System, University of Exeter** 2026-Present
- Designed a high-throughput chunk-parallel memory construction method that decouples memory formation from consolidation and supports both offline parallel and online asynchronous updates
 - Designed a workload-aware mechanism for dynamically allocating system resources to satisfy SLOs for request latency and online memory freshness
 - ★ Expected Outcome: Preparing the experimental results for submission to MLSys 2027
- GGUFxs: Lossless Compression of BF16 Tensors for LLM Inference, University of Exeter** 2026-Present
- Designed a new lossless compression file format .GGUFxs based on llama.cpp, reducing model storage by 30%
 - Implemented CPU/GPU decompression paths, achieving $\sim 5\text{-}29\times$ higher inference speed under the same GPU budget
 - ★ Expected Outcome: Preparing the experimental results for submission to EuroSys 2027
- Heterogeneous Federated Fine-Tuning System on Edge Devices, University of Exeter** 2024-2025
- Designed a heterogeneous-rank federated fine-tuning algorithm, improving accuracy by over 3% across configurations
 - Built a 10-node heterogeneous testbed using NVIDIA Jetson AGX Orin, Orin Nano, and Raspberry Pi 5
- NVP: Neural Visual Processor, UESTC & SenseTime Co., Ltd.** 2020-2022
- Designed a processor architecture for image filtering, CNNs, and stereo matching, achieving 80% PE utilization
 - Verified single-core prototype on FPGA, demonstrating $3\times$ energy efficiency improvement
 - Evaluated the PPA of an 8-core design in 40 nm CMOS, with 73.81mm^2 area and power consumption $< 2\text{ W}$
- FPGA Accelerator for MRI Segmentation, UESTC & West China Hospital of Sichuan University** 2019-2020
- Designed a parallel FPGA architecture for level-set segmentation, achieving $10.7\times$ speedup with 2.2W
 - Implemented an FPGA test demo with serial communication, supporting selection of input MRI images
- Weightel: Vehicle Intelligent Weighing System, Griffith-Elder Co., Ltd. & Camrong Co., Ltd.** 2018-2019
- Designed in-vehicle weighing and aggregation nodes for calibration and multi-node data fusion
 - Developed CAN-bus communication protocol, supporting up to 12 measurement nodes
 - Integrated and validated system in real-world environments, achieving dynamic weighing error $< 0.1\%$

PATENTS

- [1] J. Zhou, L. Chang, L. Zhou, **F. Wu**, “Data Processing Method and Apparatus, Computer Device, and Storage Medium”, PCT/CN2021/115789 (WO2022179075), China Patent CN202110221235.3 (Granted)
- [2] J. Zhou, L. Chang, L. Zhou, **F. Wu**, “Image Processing Method and Apparatus, Computer Device and Storage Medium”, PCT/CN2021/115745 (WO2022160704), China Patent CN202110132579.7 (Granted)
- [3] J. Zhou, L. Chang, L. Zhou, **F. Wu**, “Data Processing Method and Apparatus, Computer Device, and Storage Medium”, PCT/CN2021/115799 (WO2022160706), China Patent CN202110132573.X (Granted)

COMPETITIONS

National Third Prize, The 4th China Graduate Circuit Design Contest	2021
National Third Prize (Top 8.3%), The 16th China Graduate Electronics Design Contest	2021
National Second Prize (Top 3.7%), The 15th China Graduate Electronics Design Contest	2020
National Second Prize, Huawei Special Track, 15th China Graduate Electronics Design Contest	2020

ACADEMIC CONFERENCES ACTIVITIES

Poster Presenter, ICLR 2026 (Rio de Janeiro, Brazil)	04/2026
Poster Presenter, Flower AI Summit 2026 (London, UK)	04/2026
Invited Speaker, SPARKS AI Seminar, The Ohio State University (Online)	03/2026
Invited Speaker, Computer Science Research & Career Workshop (Exeter, UK)	07/2024

ACADEMIC REVIEWER

Conference	ICLR'27, AAAI'27, NeurIPS'26, ICML'26 (Gold Reviewer), ICLR'26, AAAI'26, KDD'26, ICA3PP'26, ICDCS'24, TrustCom'25, HPCC'25, ICESSE'25
Journal	IEEE TPDS, IEEE TC, IEEE IoTJ, IEEE TAI

CERTIFICATIONS & TECHNICAL SKILLS

Certifications	(1) AMD Training: Accelerating Your Application with AMD GPUs (2025) (2) NVIDIA Training: Efficient Large Language Model (LLM) Customization (2024)
Technical Skills	(1) Programming Languages: Python, C, C++, Verilog, SystemVerilog, VHDL (2) Hardware & Embedded Platforms: NVIDIA Jetson AGX Orin, Jetson Orin Nano Super, Xaiver, Raspberry Pi 5, AMD Zynq-7035, VC707, ZCU104, Atmel ATSAME51, STM32 Series (3) Frameworks & Libraries: Pytorch, Hugging Face (Transformers, PEFT, Datasets), Flower (4) Tools & Platforms: Linux (x86 & ARM), Docker, Slurm, TensorBoard, Weights & Biases, Codex, Cursor, Claude Code, PyCharm, Vivado, MATLAB, Visual Studio Code